

CUSTOMER MILESTONE BRIEF

MS #17 - Live Evidence Control Tower

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.17.0
Date	2026-08-16
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #17 - Live Evidence Control Tower converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Extends the control tower from committed demonstrations to active operational evidence and incident handling.

Why this matters to a customer

Extends the control tower from committed demonstrations to active operational evidence and incident handling.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added a canonical operational event envelope and append-only SQLite WAL reference ledger with monotonic sequence, correlation, trust, severity, time, previous-hash and content-hash fields.
- Added source observers for Factory, Portfolio, Recovery, Quality, Memory, Runtime and Audit with explicit expected latency, freshness, last observation, last identity change and trust class.
- Added resumable Server-Sent Events, bounded replay, heartbeat and live-status APIs so the local Control Tower updates without polling or manual refresh.
- Added operational alerts for dead letters, expired leases, stale runtime evidence, unavailable recovery projections and blocked release promotion, including opened/resolved ledger events.

- Upgraded the Graph Explorer into an Evidence Control Tower with connection, sequence, freshness, trust and active-alert indicators while keeping every operational surface strictly read-only.
- Added live runtime/audit snapshot reloading, portable event/status schemas, cross-platform launchers, production hardening guidance and Windows Python 3.12/3.13/3.14 discovery.
- Added event replay, chain tampering, source-change, freshness, alert, SSE and no-command-authority tests plus full verification integration.

How it differs from earlier work

MS #17 builds on MS #16 (Durable Factory Recovery Control Plane). The earlier milestone established its own bounded capability; this milestone adds live evidence control tower while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use live evidence control tower as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.17.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications

- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.